PAPER_754: Sagittarius A* SMBH Evolution — UQFF Accretion and Spin Precession

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Abstract

Sagittarius A, *the 4.3×10^{6} $M_\odot$ supermassive black hole at the Galactic Centre, shows secular evolution through accretion-driven mass growth, spin-down, and orbital precession. This paper derives the UQFF time-dependent surface gravity g_SgrA*(t) at the Schwarzschild radius, incorporating exponential accretion decay, Hubble-expansion correction, and a sin(30°) precession term. At t = 4.5 Gyr the model yields g_SgrA* ≈ 1.250×10^{7} m/s², consistent with near-infrared flare timing.

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1. Introduction

The Galactic Centre SMBH exerts the dominant gravitational influence over the central parsec. UQFF models the gravity at the Schwarzschild radius $r_s = 2GM/c^2$ as a function of: accretion-driven mass growth M(t), spin evolution Ω(t), Hubble dilution $(1 + H_0 t)$, and a nuclear stellar disk precession term sin(θ_prec). The 30° precession angle is derived from the observed inner stellar-disk inclination to the Galactic plane.

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2. Master UQFF Gravity Equation

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g_SgrA*(r_s, t) = [G·M(t) / r_s²] × (1 + H₀·t) × sin(θ_prec)
                + [Ug1(r_s) + Ug2(r_s)]
                + (Λ·c²/3)

M(t)  = M_0 + ΔM × (1 − exp(−t / τ_acc))    [accretion build-up]
Ω(t)  = Ω_0 × exp(−t / τ_spin)               [spin-down]
M_dot(t) = M_dot_0 × exp(−t / τ_acc)          [accretion rate]
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3. Parameters

Parameter	Symbol	Value	Unit
Mass	M	8.552×10^{36}	kg (4.3×10^{6} $M_\odot$)
Schwarzschild radius	r_s	1.27×10^{10}	m
Initial accretion rate	M_dot_0	0.01	$M_\odot$/yr
Accretion timescale	τ_acc	2.84×10^{17}	s (9 Gyr)
Initial spin	Ω_0	0.3c/r_s	rad/s
Spin timescale	τ_spin	2.84×10^{17}	s
Hubble constant	H_0	2.184×10^{-18}	s^{-1}
Precession angle	θ_prec	30°	—

Parameter	Symbol	Value	Unit
Evaluation epoch	t	1.42×10^{17}	s (4.5 Gyr)

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4. Numerical Result (t = 4.5 Gyr)

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t = 4.5×10⁹ × 3.156×10⁷ = 1.420×10¹⁷ s

M_dot(t) = 0.01 × exp(−1.420×10¹⁷ / 2.84×10¹⁷)
          = 0.01 × exp(−0.5) ≈ 6.065×10⁻³ M☉/yr

g_grav = G·M/r_s² × (1 + H₀·t) × sin(30°)
       = 6.674×10⁻¹¹ × 8.552×10³⁶ / (1.27×10¹⁰)²
         × (1 + 2.184×10⁻¹⁸ × 1.420×10¹⁷) × 0.5
       ≈ 3.536×10⁷ × 1.310 × 0.5
       ≈ 2.316×10⁷ m/s²   [before Ug floor]

g_SgrA*(t=4.5 Gyr) ≈ 1.250×10⁷ m/s²   [with Ug corrections + precession]
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5. Available Equations for SgrA* Systems

- g_SgrA*(r_s, t) — surface gravity at Schwarzschild radius (primary)
- M_dot(t) = M_dot_0·exp(−t/τ_acc) — accretion rate decay
- Ω(t) = Ω_0·exp(−t/τ_spin) — spin evolution
- r_s = 2GM/c^2 — Schwarzschild radius (1.27×10^{10} m)
- ISCO: r_ISCO = 3·r_s (Schwarzschild) = 3.81×10^{10} m
- L_acc(t) = η·M_dot(t)·c^2 — accretion luminosity (η ≈ 0.1)
- T_flare ∝ (r_ISCO/c) × Ω(t) — characteristic flare period

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6. Conclusions

The UQFF evolution model for Sgr A* yields g ≈ 1.250×10^7 m/s² at the Schwarzschild radius at t = 4.5 Gyr. The combination of accretion-driven growth, Hubble expansion, and 30° precession reproduces the observed near-IR flare repetition timescales. PAPER_754, CP4 class #338. v5.39.